Ruei-Che Chang

Curriculum Vitae

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EDUCATION

2020– **National Taiwan University**, *Taipei, Taiwan*.
Visiting Student at Interactive Graphics Lab.
Advisor: Bing-Yu Chen.

2019–2021 **Dartmouth College**, *New Hampshire, Hanover*.
Master of Science in Computer Science. *75% Tuition Scholarship from CS department*.
Advisor: Xing-Dong Yang.
Thesis: AccessibleCircuits: Adaptive Add-on Circuit Components for People with Blindness or Low Vision.

2014–2018 **National Cheng Kung University**, *Tainan Taiwan*.
Bachelor of Science in Electrical Engineering.

PUBLICATIONS

2021 [C.6] **Ruei-Che Chang**, Chih-An Tsao, Fang-Ying Liao, Seraphina Yong, Tom Yeh, Bing-Yu Chen, "Daedalus in the Dark: Designing for Non-Visual Accessible Construction of Laser-Cut Architecture." *In Proceedings of the ACM Symposium on User Interface Software & Technology* (**UIST'21**).

[C.5] **Ruei-Che Chang**, Wen-Ping Wang, Chi-Huan Chiang, Justin Luo, Te-Yen Wu, Zheer Xu, Bing-Yu Chen, Xing-Dong Yang, "AccessibleCircuits: Adaptive Add-On Circuit Components for People with Blindness or Low Vision." *In Proceedings of the ACM Conference on Human Factors in Computing Systems* (**CHI'21**).

[C.4] **Ruei-Che Chang**, Chi-Huan Chiang, Shuo-wen Hsu, Chih-Yun Yang, Da-Yuan Huang, Bing-Yu Chen, "TanGo: Exploring Expressive Tangible Interactions on Head-Mounted Displays." *In Proceedings of the ACM Conference on Spatial User Interface* (**SUI'20**).

2020 [C.3] Yi-Shyuan Chiang*, **Ruei-Che Chang***, Yi-Lin Chuang, Shih-Ya Chou, Hao-Ping Lee, I-Ju Lin, Jian Hua Jiang Chen, Yung-Ju Chang, "Exploring the Design Space of User-System Communication for Smart home Routine Assistants." *In Proceedings of the ACM Conference on Human Factors in Computing Systems* (**CHI'20**) (*co-primary).

[C.2] Kai-Chieh Huang, Chen-Kuo Sun, Da-Yuan Huang, Yu-Chun Chen, **Ruei-Che Chang**, Shuo-wen Hsu, Chih-Yun Yang, Bing-Yu Chen, "Glissade: Generating Balance Shifting Feedback to Facilitate Auxiliary Digital Pen Input." *In Proceedings of the ACM Conference on Human Factors in Computing Systems* (**CHI'20**).

2019 [C.1] Chi Wang, Da-Yuan Huang, Shuo-Wen Hsu, Chu-En Hou, Yeu-Luen Chiu, **Ruei-Che Chang**, Jo-Yu Lo, Bing-Yu Chen, "Masque: Exploring Lateral Skin Stretch Feedback on the Face with Head-Mounted Displays." *In Proceedings of the ACM Symposium on User Interface Software & Technology* (**UIST'19**).

RESEARCH EXPERIENCE

Jun 2020 – present **Interactive Graphics Lab, National Taiwan University**
Visiting student, advised by Bing-Yu Chen and Tom Yeh (Visiting scholar).

- Designing and conducting a study to identify accessibility challenges of laser-cut assembly. Implementing a Fusion 360 plug-in to generate tactile aids for input laser-cut model. Conducting an evaluation for tactile aids, and leading video and qualitative analysis [C.6].
- Collecting brain signal data using fNIRs on VR tasks, and analyzing data by Matlab.

- Integrating arduino and electronics, and implementing two VR applications. Designing and conducting study [C.4].

Feb 2019 – Aug 2019 ***Research Assistant, advised by Bing-Yu Chen and Da-Yuan Huang***.
Using Unity to implement six tablet applications for [C.2], and an authoring tool for designer to specify haptic feedback on a tablet [C.1].

Sep 2019 – Mar 2021 **XDiscovery Lab, Dartmouth College**
Graduate student, advised by Xing-Dong Yang.

- Designing and conducting a formative study to identify accessibility challenges, and an evaluation for our system. Designing the 3D-printed adaptions for existing circuit components using Fusion 360, and implementing a haptic-audio interface compatible to Android smartphone [C.5].
- Collecting and labeling 500+ images of breadboard circuits, and implementing a computer vision-based system using APIs (e.g., Mask-RCNN and OpenCV), and web interfaces (e.g., socket.io, MongoDB) on both student and teacher ends for remote learning.

Sep 2019 – Apr 2019 **Mobile and Ubiquitous Interaction Lab, National Yang Ming Chiao Tung University**
Research Assistant, advised by Yung-Ju (Stanley) Chang.

- Design and execute an experiment using methods, such as WoZ, speed dating, user enactment and contexual inquiry, to understand user-smart agent communication [C.3].

AWARDS

2020 **Honorable Mention Award for Glissade**, *ACM CHI 2020.*

2019 Taiwan internship scholarship from the Ministry of Education (acceptance rate 21.53%)

2017 2nd award at Taiwan Intelligent Robot Contest (30 participants)

PRESENTATION

TAICHI'21 Invited to Best of HCI session to present Daedalus and AccessibleCircuits. (TAICHI is the largest domestic HCI conference in Taiwan)

UIST'21 Daedalus in the Dark: Designing for Non-Visual Accessible Construction of Laser-Cut Architecture (held virtually).

CHI'20 AccessibleCircuits: Adaptive Add-On Circuit Components for People with Blindness or Low Vision (held virtually).

SUI'20 TanGo: Exploring Expressive Tangible Interactions on Head-Mounted Displays (held virtually).

REVIEWER

UIST '21

CHI,LBW '20 '21

SKILLS

Programming Java, C#, Python, HTML, JavaScript, ROS, OpenCV, MongoDB, LaTeX

Prototyping Arduino, 3D-printing, Fusion 360, laser-cutting

Platforms Unity3D, Android, Fusion 360